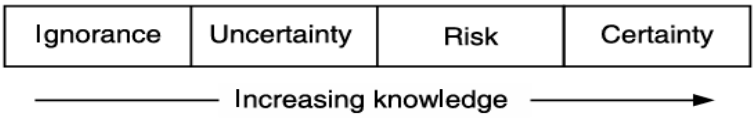
UNIT-IV

DECISION THEORY

The success or failure that an individual or organization experiences, depends to a large extent, on the ability of making acceptable decisions on time. To arrive at such a decision, a decision-maker needs to enumerate feasible and viable courses of action (alternatives or strategies), the projection of consequences associated with each course of action, and a measure of effectiveness (or an objective) to identify the best course of action.

Decision theory is both descriptive and prescriptive business modelling approach to classify the degree of knowledge and compare expected outcomes due to several courses of action. The degree of knowledge is divided into four categories: complete knowledge (i.e. certainty), ignorance, risk and uncertainty as shown



The Three Environments of Decision-Making

From a practical standpoint, the "way" you approach a decision depends entirely on the amount of information available. Decisions are made in one of three environments:

Decision Environment	Definition	Core Approach	Example
Under Certainty	**You have complete data and know exactly what the outcome of each choice will be.**	Choose the alternative that yields the highest absolute return or benefit.	**Purchasing a government bond with a fixed, guaranteed interest rate.**
Under Risk	**You do not know the exact outcome, but you know the mathematical probabilities of each scenario.**	Calculate the Expected Monetary Value (EMV) or utility by multiplying outcomes by their likelihood.	**Launching a product when market research shows a 70% chance of high success and a 30% chance of failure.**
Under Uncertainty	**You have no idea what the future holds and cannot accurately assign probabilities to potential outcomes.**	Apply logical criteria based on risk tolerance (e.g., Maximax for optimists or Minimax for pessimists).	**Investing in a highly disruptive, entirely new technology sector with zero historical data.**

Example: A company deciding how many units to produce
A company must decide whether to produce 100, 200, or 300 units of a new product.
The profit depends on market demand.

Decision	Low demand	Medium demand	High demand
Produce 100 units	₹40,000	₹40,000	₹40,000
Produce 200 units	₹20,000	₹70,000	₹80,000
Produce 300 units	–₹20,000	₹50,000	₹1,20,000

Now the three decision situations can be explained using this same example.

1. Decision-making under Certainty

Meaning

Under **certainty**, the decision-maker **knows exactly which state of nature will occur**.

Example

Suppose the company has reliable confirmed orders showing that the demand will be **high**.

Therefore, the company knows:

High demand will occur with certainty.

Looking at the table:

- Produce 100 → ₹40,000
- Produce 200 → ₹80,000
- Produce 300 → ₹1,20,000

The company can choose **300 units**, because the future state is known.

Key point: Certainty = State of nature is known

2. Decision-making under Risk

Meaning

Under **risk**, the possible future states are known **and their probabilities are known**.

Suppose market research gives:

Demand	Probability
Low	0.20
Medium	0.50
High	0.30

Now calculate the **Expected Monetary Value (EMV)**.

Produce 100

EMV (100) =0.20(40,000) +0.50(40,000) +0.30(40,000) = ₹40,000

Produce 200

EMV (200) =0.20(20,000) +0.50(70,000) +0.30(80,000) =₹59,000

Produce 300

EMV (300) =0.20(−20,000) +0.50(50,000) +0.30(1,20,000) = ₹46,000

So the EMVs are:

Decision	EMV
100 units	₹40,000
200 units	**₹59,000**
300 units	₹46,000

Under the **EMV criterion**, the decision corresponding to ₹59,000 has the highest expected monetary value.

Key point: Risk = Possible outcomes known + probabilities known

3. Decision-making under Uncertainty

Meaning

Under **uncertainty**, the possible states of nature may be known, but their **probabilities are not known or cannot be reliably estimated**.

Suppose the company knows that demand could be:

- Low
- Medium
- High

But it has **no reliable information about the probabilities**.

So the company cannot say:

"Low demand has probability 0.20." or "High demand has probability 0.30."

Therefore, EMV cannot be calculated reliably.

The company must use decision criteria such as:

- **Maximax criterion** – optimistic approach
- **Maximin criterion** – conservative approach
- **Minimax Regret criterion**
- **Laplace criterion** – assume equal likelihood when appropriate

Key point: Uncertainty = Possible outcomes known, probabilities unknown

DECISION MAKING THEORY

1. INTRODUCTION TO DECISION MAKING

1.1 Meaning of Decision Making

Decision making is the process of selecting the most appropriate course of action from a set of available alternatives to achieve a desired objective.

In real-life situations, a decision-maker may not know exactly what will happen in the future. Different future conditions may lead to different outcomes. Decision theory provides mathematical and logical tools for choosing among alternatives.

Simple example

A company wants to decide whether to develop a new software product.

It has three alternatives:

- Develop a small system
- Develop a medium system
- Develop a large system

The profit will depend on future market demand:

- Low demand
- Medium demand
- High demand

The company must decide which alternative to choose.

2. Importance of Decision Making

Decision making is important because organizations and individuals frequently have to make choices when resources are limited and future outcomes are uncertain.

Major applications

1. **Resource allocation**
    - How much money, time, computing power, or manpower should be allocated?
2. **Project selection**
    - Which software or engineering project should be undertaken?
3. **Production planning**
    - How many units should be produced?
4. **Investment decisions**
    - Which investment option should be selected?
5. **Technology selection**

- Which hardware, software, cloud platform, or technology should be adopted?

6. **Capacity planning**
    - How much server capacity should be installed?
7. **Risk management**
    - What action should be taken when future outcomes are uncertain?
8. **Artificial intelligence and machine learning**
    - Selecting an action based on predicted outcomes and associated costs.

3. Elements of a Decision Problem

A decision problem generally consists of four important elements:

Actions + States of Nature + Payoffs + Decision Criterion

The three fundamental elements are **actions, states of nature, and payoffs**.

3.1 Actions / Alternatives

An **action** is a possible decision or course of action available to the decision-maker.

They are usually denoted by A1, A2, …, Am

Example

A software company has three possible actions:

A1=Develop basic software

A2=Develop standard software

A3=Develop advanced software

The company must select one of these actions.

3.2 States of Nature

A **state of nature** represents a possible future condition that affects the outcome of a decision but is **not controlled by the decision-maker**.

They are usually denoted by S1, S2, … …, Sn

For example: S1=Low demand

S2=Medium demand

S3=High demand

The company cannot control whether demand will be low, medium, or high.

Important distinction

Action → controlled by decision-maker

State of nature → not controlled by decision-maker

3.3 Payoff

A **payoff** is the result obtained when a particular action is chosen and a particular state of nature occurs.

Payoffs may represent:

- Profit
- Revenue
- Cost savings
- Performance
- Production
- Time saved

For a profit problem, a higher payoff is generally preferred.

For a cost problem, a lower payoff is generally preferred.

4. Decision Table

A **decision table** presents the actions, states of nature, and corresponding payoffs in a systematic form.

Example

A company has three production alternatives.

Action	Low demand	Medium demand	High demand
A1: Small plant	₹40,000	₹40,000	₹40,000
A2: Medium plant	₹20,000	₹70,000	₹80,000
A3: Large plant	–₹20,000	₹50,000	₹1,20,000

Here:

- **Actions:** Small, Medium, Large plant
- **States of nature:** Low, Medium, High demand
- **Entries:** Payoffs

This table is called a **payoff table** or **payoff matrix**.

5. Payoff Matrix

A **payoff matrix** is a mathematical representation of the outcomes associated with different actions and states of nature.

In general: $P = [P_{ij}]$

Where P_{ij} = payoff obtained by choosing A_i when state S_j occurs.

For example: $$P = \begin{bmatrix} 40000 & 40000 & 40000 \\ 20000 & 70000 & 80000 \\ -20000 & 50000 & 120000 \end{bmatrix}$$

Rows represent **actions** and columns represent **states of nature**.

6. Types of Decision Making

Decision problems are commonly classified into:

1. Decision making under certainty

The state of nature is known with certainty.

2. Decision making under risk

The possible states are known and their probabilities are known.

3. Decision making under uncertainty

The possible states are known, but their probabilities are unknown.

This chapter focuses particularly on **uncertainty and risk**.

PART A: DECISION MAKING UNDER UNCERTAINTY

Meaning of Decision Making Under Uncertainty

A decision is made under **uncertainty** when:

1. There are several possible states of nature.
2. The decision-maker does not know which state will occur.
3. The probabilities of the states are **not known**.

For example, a company is launching a completely new product and has no reliable information about future demand.

Possible demand:

- Low
- Medium
- High

But probabilities cannot be estimated reliably.

Therefore, probability-based methods such as EMV cannot directly be applied.

Instead, several **decision criteria** are used.

Decision Criteria Under Uncertainty

The major criteria are:

1. **Optimism (Maximax or Minimin) criterion**
2. **Pessimism (Maximin or Minimax) criterion**
3. **Minimax Regret criterion**
4. **Laplace criterion** (Equal probabilities)
5. **Hurwicz criterion** (Coefficient of optimism)

We will use the following payoff matrix throughout example:

Alternative	Low demand	Medium demand	High demand
A1 Small	40	40	40
A2 Medium	20	70	80
A3 Large	–20	50	120

Payoffs are in ₹ **thousands**.

1. Maximax Criterion

Maximax means: Maximum of the maximum payoffs

It represents an **optimistic approach**.

The decision-maker assumes that the most favourable state of nature will occur.

Step 1: Find the maximum payoff for each action

Action	Payoffs	Row maximum
A1	40, 40, 40	40
A2	20, 70, 80	80
A3	–20, 50, 120	**120**

Step 2

Select the largest row maximum:

Max (40, 80, 120) = 120

Therefore, according to **Maximax**, select: A3

Interpretation

The decision-maker focuses on the **best possible payoff**.

Maximax = Best of the best

2. Maximin Criterion

Maximin means: Maximum of the minimum payoffs

It represents a **conservative approach**.

The decision-maker considers the worst outcome for each action and then selects the action whose worst outcome is the best among the alternatives.

Step 1: Find minimum payoff in each row

Action	Payoffs	Row minimum
A1	40, 40, 40	**40**
A2	20, 70, 80	20
A3	–20, 50, 120	–20

Step 2

Select the largest minimum: max (40, 20, -20) = 40

Therefore: A1

Interpretation

The decision-maker asks: "If the worst possible outcome occurs, which decision protects me best?"

Maximin = Best of the worst

Maximax vs Maximin

Criterion	Approach	Calculation
Maximax	Optimistic	Maximum of row maxima
Maximin	Conservative	Maximum of row minima

For the example:

Maximax=A3 Maximin=A1

The different answers occur because the two criteria have different attitudes toward uncertainty.

3. Minimax Regret Criterion

Regret is the loss caused by not selecting the best action after the actual state of nature becomes known.

It is also called:

- Opportunity loss
- Regret

For each state of nature: Regret = Best payoff in that state−Actual payoff

Constructing the Regret Matrix

Original payoff matrix:

Action	Low	Medium	High
A1	40	40	40
A2	20	70	80
A3	–20	50	120

Step 1: Find the maximum payoff in each column

- Low demand → 40
- Medium demand → 70
- High demand → 120

Step 2: Calculate regret

For Low: 40−40=0 40−20=20 40−(−20) = 60

For Medium: 70−40=30 70−70=0 70−50=20

For High: 120−40=80 120−80=40 120−120=0

Therefore:

Action	Low	Medium	High	Maximum regret
A1	0	30	80	**80**
A2	20	0	40	**40**
A3	60	20	0	**60**

Step 3: Apply Minimax Regret

Choose the **minimum of the maximum regrets**: min (80, 40, 60) = 40

Therefore: A2

Minimax Regret = Minimize the maximum possible regret.

4. Laplace Criterion

Since the probabilities of states of nature are not known, it is assumed that all states of nature will occur with equal probability, i.e. each state of nature is assigned an equal probability. As states of nature are mutually exclusive and collectively exhaustive, so the probability of each of these must be: 1/(number of states of nature). The working method is summarized as follows:

(a) Assign equal probability value to each state of nature by using the formula: $\frac{1}{number\ of\ states\ of\ nature}$

(b) Compute the expected (or average) payoff for each alternative (course of action) by adding all the payoffs and dividing by the number of possible states of nature, or by applying the formula:

(Probability of state of nature j) × (Payoff value for the combination of alternative i

and state of nature j.)

(c) Select the best expected payoff value (maximum for profit and minimum for cost).

Meaning: The **Laplace criterion** is used when probabilities are unknown and the decision-maker assumes that all states of nature are **equally likely**.

If there are n states $P(S_i) = \frac{1}{n}$

The average payoff is $L(A_i) = \frac{\sum_{j=1}^{n} P_{ij}}{n}$

Choose the action with the **highest average payoff** for a payoff-maximization problem.

Example

Using our payoff matrix:

$$L(A_1) = \frac{40+40+40}{3} = 40$$

$$L(A_2) = \frac{20+70+80}{3} = 56.67$$

$$L(A_3) = \frac{-20+50+120}{3} = 50$$

Therefore: A2, Because 56.67 > 50 > 40

Interpretation: Laplace says: "Since I have no probability information, treat the states equally."

Important note: Equal probabilities are an **assumption of the Laplace criterion**, not evidence that the states are actually equally likely.

5. Hurwicz Criterion

This criterion suggests that a decision-maker should be neither completely optimistic nor pessimistic and, therefore, must display a mixture of both. Hurwicz, who suggested this criterion, introduced the idea of a coefficient of optimism (denoted by α) to measure the decision-maker's degree of optimism. This coefficient lies between 0 and 1, where 0 represents a complete pessimistic attitude about the future and 1 a complete optimistic attitude about the future. Thus, if α is the coefficient of optimism, then (1 – α) will represent the coefficient of pessimism.

The Hurwicz approach suggests that the decision-maker must select an alternative that maximizes

H (Criterion of realism) = α (Maximum payoff) + (1 – α) (Minimum payoff)

The working method is summarized as follows:

a) Decide the coefficient of optimism α (alpha) and then coefficient of pessimism (1 – α).
b) For each decision alternative select the largest and lowest payoff value and multiply these with α and (1 – α) values, respectively. Then calculate the weighted average, H by using above formula.
c) Select an alternative with best weighted average payoff value.

Meaning: The **Hurwicz criterion** combines optimistic and pessimistic viewpoints.

Practice Problems:

	States of Nature		
Strategies	N_1	N_2	N_3
S_1	7,00,000	3,00,000	1,50,000
S_2	5,00,000	4,50,000	0
S_3	3,00,000	3,00,000	3,00,000

Which strategy should the concerned executive choose on the basis of

(a) Maximin criterion
(b) Maximax criterion
(c) Minimax regret criterion
(d) Laplace criterion?

1. The following matrix gives the payoff (in Rs) of different strategies (alternatives) S_1, S_2 and S_3 against conditions (events) N_1, N_2, N_3 and N_4.

Strategy	*State of Nature*			
	N_1	N_2	N_3	N_4
S_1	4,000	− 100	6,000	18,000
S_2	20,000	5,000	400	0
S_3	20,000	15,000	− 2,000	1,000

Indicate the decision taken under the following approaches: (i) Pessimistic, (ii) Optimistic, (iii) Equal probability, (iv) Regret, (v) Hurwicz criterion, the degree of optimism being 0.7.

2. In a toy manufacturing company, suppose the product acceptance probabilities are not known but the following data is known:

Product Acceptance	*Anticipated First Year Profit ('000 Rs) Product Line*		
	Full	*Partial*	*Minimal*
Good	8	70	50
Fair	50	45	40
Poor	− 25	− 10	0

Determine the optimal decision under each of the following decision criteria and show how you arrived at it: (a) Maximax, (b) Maximin, (c) Equal likelihood and (d) Minimax regret?

3. The following is a payoff (in rupees) table for three strategies and two states of nature:

Strategy	*State of Nature*	
	N_1	N_2
S_1	40	60
S_2	10	− 20
S_3	− 40	150

Select a strategy using each of the following decision criteria: (a) Maximax, (b) Minimax regret, (c) Maximin, (d) Minimum risk, assuming equiprobable states.

4. Mr Sethi has Rs 10,000 to invest in one of three options: A, B or C. The return on his investment depends on whether the economy experiences inflation, recession, or no change at all. The possible returns under each economic condition are given below:

Strategy	*State of Nature*		
	Inflation	*Recession*	*No Change*
A	2,000	1,200	1,500
B	3,000	800	1,000
C	2,500	1,000	1,800

What should he decide, using the pessimistic criterion, optimistic criterion, equally likely criterion and regret criterion?

1. (i) S_2, (ii) S_2 or S_3, (iii) S_3, (iv) S_1, (v) S_2
2. (a) Full, (b) Minimal, (c) Full or partial, (d) Partial
3. (a) S_3; Rs 150 (b) S_3; Rs 80 (c) S_1; Rs 40 (d) S_3; Rs 55
4. Choose A: Rs 120, Choose B: Rs 300, Choose C: Rs 176.6, Choose C: Rs 50

PART B: DECISION MAKING UNDER RISK

Meaning of Decision Making Under Risk

Decision making under **risk** occurs when:

1. The possible states of nature are known.
2. The probabilities of those states are known or can be estimated.

For example:

Demand	Probability
Low	0.20
Medium	0.50
High	0.30

Here:

0.20 + 0.50 + 0.30 = 1

Because probabilities are available, we can use **EMV** and **EOL**.

Expected Monetary Value (EMV)

Definition:

Expected Monetary Value is the probability-weighted average of the payoffs associated with a decision.

The formula is: $EMV(A_i) = \sum_{j=1}^{n} P(S_j)P_{ij}$

where:

- P(Sj) = probability of state Sj
- Pij = payoff of action Ai

For a profit problem: Choose the alternative with maximum EMV

Worked Example: EMV

Consider:

Action	Low	Medium	High
A1	40	40	40
A2	20	70	80
A3	–20	50	120

Probabilities: P(L) = 0.20, P(M) = 0.50, P(H) = 0.30

EMV of A1 EMV (A1) = 0.2 (40) + 0.5 (40) + 0.3 (40) = 40

EMV of A2 EMV (A2) = 0.2 (20) + 0.5 (70) + 0.3 (80) = 63

EMV of A3 EMV (A3) = 0.2 (−20) + 0.5 (50) + 0.3 (120) = 57

Therefore:

Alternative	EMV
A1	₹40,000
A2	**₹63,000**
A3	₹57,000

Under the EMV criterion, A2 has the highest expected monetary value.

Expected Opportunity Loss (EOL)

Meaning: Expected Opportunity Loss is the expected value of regret.

First construct the **regret/opportunity-loss matrix**.

Recall:

Action	Low	Medium	High
A1	0	30	80
A2	20	0	40
A3	60	20	0

The probabilities are: 0.20, 0.50, 0.30

EOL of A1, EOL (A1) = 0.2 (0) + 0.5 (30) + 0.3 (80) = 0 + 15 + 24 = 39

EOL of A2, EOL (A2) = 0.2 (20) + 0.5 (0) + 0.3 (40) = 4 + 0 + 12 = 16

EOL of A3, EOL (A3) = 0.2 (60) + 0.5 (20) + 0.3 (0) = 12 + 10 + 0 = 22

Therefore:

Alternative	EOL
A1	₹39,000
A2	**₹16,000**
A3	₹22,000

For EOL:

Choose the alternative with minimum EOL

Thus, A2 has the minimum EOL.

Important Relationship Between EMV and EOL

A very important result in decision theory is:

Min (EOL) = EVPI− Max (EMV)

where **EVPI** means **Expected Value of Perfect Information**.

For our example:

Max EMV = 63

Expected payoff with perfect information:

EVPI payoff = 0.2 (40) + 0.5 (70) + 0.3 (120) = 8 + 35 + 36 = 79

Therefore: EVPI = 79 – 63 = 16

Notice: Min EOL = 16

Therefore: EVPI = Min EOL

This provides a useful check on calculations.

EMV vs EOL

EMV	EOL
Expected Monetary Value	Expected Opportunity Loss
Uses payoff matrix	Uses regret matrix
Higher is preferred	Lower is preferred
Measures expected payoff	Measures expected regret
Formula uses payoffs	Formula uses opportunity losses

Problem:

A company is considering three server configurations for a new computing application.

The future workload may be:

- Low
- Medium
- High

The estimated annual profit (₹ lakhs) is:

Server	Low workload	Medium workload	High workload
S1: Basic	10	10	10
S2: Standard	6	18	24
S3: High-performance	–4	14	36

Assume probabilities: P(L) = 0.3, P(M) = 0.4, P(H) = 0.3

Solution:

(a) Maximax

Row maxima: S1=10, S2=24, S3=36

Therefore: S3

(b) Maximin

Row minima: S1=10, S2=6, S3=−4

Therefore: S1

(c) Minimax Regret

Column maxima: 10, 18, 36

Regret matrix:

Server	Low	Medium	High	Maximum regret
S1	0	8	26	26
S2	4	0	12	**12**
S3	14	4	0	14

Therefore: S2

(d) Laplace

L(S1) = (10 + 10 + 10)/3 = 10

L(S2) = (6 + 18 + 24)/3 = 16

L(S3) = (−4 + 14 + 36)/ 3=15.33

Therefore: S2

(e) Hurwicz

Suppose: α=0.7

For S1: H(S1) = 0.7 (10) + 0.3 (10) = 10

For S2: H(S2) = 0.7 (24) + 0.3 (6) = 18.6

For S3: H(S3) = 0.7 (36) + 0.3 (−4) = 24

Thus: S3

(f) EMV

EMV(S1) = 0.3 (10) + 0.4 (10) + 0.3 (10) = 10

EMV(S2) = 0.3 (6) + 0.4 (18) + 0.3 (24) = 1.8 + 7.2 + 7.2 = 16.2

EMV(S3) = 0.3 (−4) + 0.4 (14) + 0.3 (36) = −1.2 + 5.6 + 10.8 = 15.2

Therefore: S2

Practice Problems

Problem 1 — Maximax and Maximin

A company has the following payoff matrix, in ₹ lakhs:

Alternative	Low	Medium	High
A	30	30	30
B	10	50	70
C	−20	40	100

Find the decision using:

1. Maximax
2. Maximin

Answer:

Row maxima: 30, 70, 100

Maximax: C

Row minima: 30, 10, −20

Maximin: A

Problem 2 — Minimax Regret

Alternative	Low	Medium	High
A	50	50	50
B	30	80	100
C	0	60	150

Construct the regret matrix and find the decision using **Minimax Regret**.

Answer: Column maxima: 50, 80, 150

Regret matrix:

Alternative	Low	Medium	High	Maximum
A	0	30	100	100
B	20	0	50	**50**
C	50	20	0	**50**

Minimum maximum regret: **50**

There is a **tie between B and C** under the Minimax Regret criterion.

Problem 3 — Laplace Criterion

Alternative	State 1	State 2	State 3	State 4
A	40	50	60	30
B	20	80	70	40
C	60	40	50	70

Find the decision using the **Laplace criterion**.

Solution

L (A) = (40 + 50 + 60 + 30) / 4 = 45

L (B) = (20 + 80 + 70 + 40) / 4 = 52.5

L (C) = (60 + 40 + 50 + 70) / 4 = **55**

Therefore: C

Problem 4 — Hurwicz Criterion

Consider:

Alternative	Low	High
A	20	60
B	40	70
C	–10	100

If the coefficient of optimism is α = 0.6

find the decision using the Hurwicz criterion.

Answer

H (A) = 0.6 (60) + 0.4 (20) = 44

H (B) = 0.6 (70) + 0.4 (40) = 58

H (C) = 0.6 (100) + 0.4 (−10) = 56

Therefore: B

Problem 5 — Expected Monetary Value

A software company has three deployment strategies.

Strategy	Low demand	Medium demand	High demand
A	20	20	20
B	10	50	80
C	–30	60	120

Probabilities: P(L) = 0.2, P(M) = 0.5, P(H) = 0.3

Find the EMV for each strategy and determine the strategy with maximum EMV.

Answer

EMV(A) =0.2 (20) + 0.5 (20) + 0.3 (20) = 4 +10+ 6 = 20

EMV(B) = 0.2 (10) + 0.5 (50) + 0.3 (80) = 2 + 25 + 24 = 51

EMV(C) = 0.2 (−30) + 0.5 (60) + 0.3 (120) = −6 + 30 + 36 = 60

Thus: EMV (C) = 60 is the highest.

Problem 6 — Expected Opportunity Loss (EOL)

Using Problem 5, construct the opportunity-loss matrix and calculate EOL.

Step 1: Column maxima: 20, 60, 120

Step 2: Regret matrix

Strategy	Low	Medium	High
A	0	40	100
B	10	10	40
C	50	0	0

Step 3: EOL

EOL(A) = 0.2 (0) + 0.5 (40) + 0.3 (100) = 50

EOL(B) = 0.2 (10) + 0.5 (10) + 0.3 (40) = 19

EOL(C) = 0.2 (50) +0.5 (0) + 0.3 (0) = **10**

Therefore: C, has the minimum EOL.

Notice: Max EMV = 60

And

EVPI = 70 – 60 = 10

which agrees with: Min EOL = 10

Problem 7: The manager of a flower shop promises its customers delivery within four hours on all flower orders. All flowers are purchased on the previous day and delivered to Parker by 8.00 am the next morning. The daily demand for roses is as follows.

Dozens of roses	70	80	90	100
Probability	0.1	0.2	0.4	0.3

The manager purchases roses for Rs 10 per dozen and sells them for Rs 30. All unsold roses are donated to a local hospital. How many dozens of roses should Parker order each evening to maximize its profits? What is the optimum expected profit?

Solution:

This is a **decision-making under risk**. We can solve it using the **Expected Monetary Value (EMV)** approach.

Given data

Demand (dozens)	70	80	90	100
Probability	0.1	0.2	0.4	0.3

- Purchase cost = **₹10 per dozen**
- Selling price = **₹30 per dozen**
- Unsold roses are donated → **salvage value = ₹0**

Therefore,

Profit = 30 (roses sold) – 10 (roses ordered)

Since the manager must decide the order **before knowing demand**, consider possible order quantities: **70, 80, 90, 100 dozen**.

Profit Table:

Order quantity	Demand 70	Demand 80	Demand 90	Demand 100
70	₹1400	₹1400	₹1400	₹1400
80	₹1300	₹1600	₹1600	₹1600
90	₹1200	₹1500	₹1800	₹1800
100	₹1100	₹1400	₹1700	₹2000

For example, if **90 dozen are ordered** and demand is 80:

Profit = 80 (30) – 90 (10) = 2400 − 900 = ₹1500

Expected Profit:

If 70 dozen are ordered: EMV = 1400 (0.1 + 0.2 + 0.4 + 0.3) = ₹1400

If 80 dozen are ordered: EMV = 1300 (0.1) + 1600 (0.2 + 0.4 + 0.3) = 130 + 1440 = ₹1570

If 90 dozen are ordered:

EMV = 1200 (0.1) + 1500 (0.2) + 1800 (0.4) + 1800 (0.3) = 120+300+720+540 = **₹1680**

If 100 dozen are ordered:

EMV = 1100 (0.1) + 1400 (0.2) + 1700 (0.4) + 2000 (0.3) = 110+280+680+600 = ₹1670

Therefore, **the manager should order**: **90 dozen** and the **optimum expected profit is** ₹1,680

Decision: Order 90 dozen roses every evening.

Problem 8: A retailer purchases cherries every morning at Rs 50 a case and sells them for Rs 80 a case. Any case that remains unsold at the end of the day can be disposed of the next day at a salvage value of Rs 20 per case (thereafter they have no value). Past sales have ranged from 15 to 18 cases per day. The following is the record of sales for the past 120 days.

Cases sold	15	16	17	18
Number of days	12	24	48	36

Find out how many cases should the retailer purchase per day in order to maximize his profit.

Solution: This is a **decision-making under risk / inventory problem**. Since the past 120 days provide probabilities of different demand levels, we can use the **Expected Monetary Value (EMV)** method.

1. Given data

- Purchase cost = **₹50 per case**
- Selling price = **₹80 per case**
- Salvage value = **₹20 per unsold case**
- Number of past days = **120**

Cases sold (Demand)	**15**	**16**	**17**	**18**
Number of days	12	24	48	36
Probability	12/120=0.10	24/120=0.20	48/120=0.40	36/120=0.30

Thus,

P (15) = 0.10, P (16) = 0.20, P (17) = 0.40, P (18) = 0.30

2. Construct the profit table

The profit formula is, Profit = 80 (cases sold) + 20 (unsold cases) – 50 (cases purchased)

We consider purchasing **15, 16, 17, or 18 cases**.

If 15 cases are purchased

Even if demand is 16, 17, or 18, only 15 cases can be sold.

Profit = 15 (80) – 15 (50) = ₹ 450 so profit is ₹450 in every situation.

If 16 cases are purchased

- Demand = 15: 15 (80) + 1 (20) −16 (50) = ₹420
- Demand ≥ 16: 16 (80) – 16 (50) = ₹480

If 17 cases are purchased

- Demand = 15: 15 (80) +2 (20) – 17 (50) = ₹390
- Demand = 16: 16 (80) + 1 (20) – 17 (50) = ₹450

- Demand ≥ 17: 17 (80) – 17 (50) = ₹510

If 18 cases are purchased

- Demand = 15: 15 (80) + 3 (20) – 18 (50) = ₹360
- Demand = 16: 16 (80) + 2 (20) – 18 (50) = ₹390
- Demand = 17: 17 (80) + 1 (20) − 18 (50) = ₹480
- Demand = 18: 18 (80) – 18 (50) = ₹540

Therefore, the **profit matrix** is:

Purchase / Demand	15	16	17	18
15 cases	450	450	450	450
16 cases	420	480	480	480
17 cases	390	450	510	510
18 cases	360	420	480	540

3. Calculate EMV

Purchase 15 cases

EMV (15) = 450 (0.10) + 450 (0.20) + 450 (0.40) + 450 (0.30) = 45 +90 + 180+ 135 = ₹ 450

Purchase 16 cases

EMV (16) = 420 (0.10) + 480 (0.20) + 480 (0.40) + 480 (0.30) = 42 + 96 + 192 + 144 = ₹ 474

Purchase 17 cases

EMV (17) = 390 (0.10) + 450 (0.20) + 510 (0.40) + 510 (0.30) = 39 + 90 + 204 + 153 = **₹ 486**

Purchase 18 cases

EMV (18) = 360 (0.10) + 420 (0.20) + 480 (0.40) +540 (0.30) = 36 + 84 + 192 + 162 = ₹ 474

Since the **highest EMV of Rs 486** corresponds to the course of action 17, the retailer **must purchase 17 cases of cherries every morning.**

Problem 9:

You are given the following payoffs of three Acts A1, A2 and A3 and the events E1, E2, E3.

States of Nature	Three Acts		
	A1	A2	A3
E1	25	–10	–125
E2	400	440	400
E3	650	740	750

The probabilities of the states of nature are 0.1, 0.7 and 0.2, respectively. Calculate and tabulate the EMV and conclude which would prove to be the best course of action.

Solution:

Selection of the Best Act Using EMV

There are three possible **Acts** A1, A2, A3 and three **States of Nature** E1, E2, E3.

The probabilities are: P (E1) = 0.1, P (E2) = 0.7, P (E3) = 0.2

The payoff table is:

	Three Acts			**Probability**
State of Nature	**A1**	**A2**	**A3**	
E1	25	-10	-125	**0.1**
E2	400	440	400	**0.7**
E3	650	740	750	**0.2**

Step 1: EMV of A1

The formula is

EMV (Ai) = ∑P(Ej) × Payoff (Ai, Ej)

Therefore,

EMV (A1) = 0.1 (25) + 0.7 (400) + 0.2 (650) = 2.5 + 280 + 130 = 412.5

Step 2: EMV of A2

EMV (A2) = 0.1 (−10) + 0.7 (440) + 0.2 (740) = −1 + 308 + 148 = **455**

Step 3: EMV of A3

EMV (A3) = 0.1 (−125) + 0.7 (400) + 0.2 (750) = −12.5 + 280 + 150 = 417.5

According to the EMV criterion, select the act having the **maximum expected monetary value**.

Max (412.5, 455, 417.5) = **455**

Answer: Choose A2 because, EMV (A2) = 455 is the highest expected payoff.

Problem 10: The management of a company is faced with the problem of choosing one of three products that it wants to manufacture. The potential demand for each product may turn out to be good, moderate or poor. The probabilities for each of the states of nature were estimated as follows.

Product	*Nature of Demand*		
	Good	*Moderate*	*Poor*
X	0.70	0.20	0.10
Y	0.50	0.30	0.20
Z	0.40	0.50	0.10

The estimated profit or loss in rupees under the three states may be taken as:

Product	*Good*	*Moderate*	*Poor*
X	30,000	20,000	10,000
Y	60,000	30,000	20,000
Z	40,000	10,000	– 15,000

Prepare the expected value table, and advise the management about the choice of product.

Solution:

Choice of Product Using Expected Value

The company has to choose one of three products: X, Y, Z

The possible states of demand are:

- Good
- Moderate
- Poor

The important point here is that **the probabilities are different for each product**.

Given probability table

Product	Good	Moderate	Poor
X	0.70	0.20	0.10
Y	0.50	0.30	0.20
Z	0.40	0.50	0.10

Payoff table

Product	Good	Moderate	Poor
X	₹30,000	₹20,000	₹10,000
Y	₹60,000	₹30,000	₹20,000
Z	₹40,000	₹10,000	-₹15,000

We calculate the expected value for each product.

Step 1: Expected Value of Product X

EV (X) = 0.70 (30000) + 0.20 (20000) + 0.10 (10000) = 21000 + 4000 + 1000 = ₹ 26,000

Step 2: Expected Value of Product Y

EV (Y) = 0.50 (60000) + 0.30 (30000) + 0.20 (20000) = 30000 + 9000 + 4000 = ₹ 43,000

Step 3: Expected Value of Product Z

EV (Z) = 0.40 (40000) + 0.50 (10000) + 0.10 (−15000) = 16000 + 5000 – 1500 = ₹ 19,500

Expected Value Table

Product	Good	Moderate	Poor	Expected Value
X	₹21,000	₹4,000	₹1,000	₹26,000
Y	₹30,000	₹9,000	₹4,000	**₹43,000**
Z	₹16,000	₹5,000	-₹1,500	₹19,500

Therefore,

EV (Y) > EV (X) > EV (Z)

and the maximum expected value is ₹ 43,000

Answer: The management should choose Product Y according to the Expected Value criterion.

The expected monetary return associated with this decision is: ₹ 43,000.

Example 11.8 The probability of demand for hiring cars on any day in a given city is as follows:

No. of cars demanded	:	0	1	2	3	4
Probability	:	0.1	0.2	0.3	0.2	0.2

Cars have a fixed cost of Rs 90 each day to keep the daily hire charges (variable costs of running) Rs 200. If the car-hire company owns 4 cars, what is its daily expectation? If the company is about to go into business and currently has no car, how many cars should it buy?

Solution Given that Rs 90 is the fixed cost and Rs 200 is variable cost. The payoff values with 4 cars at the disposal of decision-maker are calculated as under:

No. of cars demanded	:	0	1	2	3	4
Payoff (with 4 cars)	:	$0 - 90 \times 4 = -360$	$200 - 90 \times 4 = -160$	$400 - 90 \times 4 = 40$	$600 - 90 \times 4 = 240$	$800 - 90 \times 4 = 440$

Thus, the daily expectation is obtained by multiplying the payoff values with the given corresponding probabilities of demand:

$$\text{Daily Expectation} = (-360)(0.1) + (-160)(0.2) + (40)(0.3) + (240)(0.2) + (440)(0.2) = \text{Rs } 80$$

The conditional payoffs and expected payoffs for each course of action are shown in Tables 11.6 and 11.7.

Demand of Cars	*Probability*	*Conditional Payoff (Rs) due to Decision to Purchase Cars (Course of Action)*				
		0	*1*	*2*	*3*	*4*
0	0.1	0	– 90	– 180	– 270	– 360
1	0.2	0	110	20	– 70	– 160
2	0.3	0	110	220	130	40
3	0.2	0	110	220	330	240
4	0.2	0	110	220	330	440

Demand of Cars	*Probability*	*Conditional Payoff (Rs) due to Decision to Purchase Cars (Course of Action)*				
		0	*1*	*2*	*3*	*4*
0	0.1	0	– 9	– 18	– 27	– 36
1	0.2	0	22	4	– 14	– 32
2	0.3	0	33	66	39	12
3	0.2	0	22	44	66	48
4	0.2	0	22	44	66	88
EMV		0	90	**140**	130	80

Since the EMV of Rs 140 for the course of action 2 is the highest, the company should buy 2 cars.

Expected Opportunity Loss (EOL)

Expected opportunity loss (EOL), also called expected value of regret, is an alternative decision criterion for decision making under risk. The EOL is defined as the difference between the highest profit (or payoff) and the actual profit due to choosing a particular course of action in a particular state of nature. Hence, EOL is the amount of payoff that is lost by not choosing a course of action resulting to the minimum payoff in a particular state of nature. A course of action resulting to the minimum EOL is preferred. Mathematically, EOL is stated as follows.

$$EOL\ (State\ of\ nature, N_i) = \sum_{i=1}^{m} l_{ij} P_i$$

l_{ij} = opportunity loss due to state of nature, N_i and course of action, S_j

P_i = probability of occurrence of state of nature, N_i

The Procedure:

1. Prepare a conditional payoff values matrix for each combination of course of action and state of nature along with the associated probabilities.
2. For each state of nature calculate the conditional opportunity loss (COL) values by subtracting each payoff from the maximum payoff.
3. Calculate the EOL for each course of action by multiplying the probability of each state of nature with the COL value and then adding the values.
4. Select a course of action for which the EOL is minimum.

Problem 1: A newspaper boy purchases magazines at Rs 3 each and sells them at Rs 5 each. He cannot return the unsold magazines. The probability distribution of the demand for the magazine is given below.

Demand	16	17	18	19	20
probabilities	0.10	0.15	0.20	0.25	0.30

Determine how many copies of magazine should he purchase (Apply EOL criterion).

Solution: This is a newsvendor/inventory decision problem. Since the question specifically asks to apply the Expected Opportunity Loss (EOL) criterion, we first construct the opportunity loss (regret) table, then calculate EOL for each possible purchase quantity.

Given data

- Purchase cost = ₹3 per copy
- Selling price = ₹5 per copy
- No return/salvage value for unsold copies

Demand	**16**	**17**	**18**	**19**	**20**
Probability	0.10	0.15	0.20	0.25	0.30

The possible purchase quantities are: 16, 17, 18, 19, 20

1. Construct the Opportunity Loss Table

Case 1: Demand = 16

If we purchase exactly 16:

Profit = 16 (5) – 16 (3) = **₹ 32** This is the maximum profit when demand is 16.

If we purchase 17, one copy is unsold, causing a loss of ₹3.

Case 2: Demand = 17

Maximum profit occurs when 17 copies are purchased:

17 (5) – 17 (3) = **₹ 34** If only 16 are purchased, one customer cannot be served.

Lost profit: 5 – 3 = ₹ 2

		Demand				
		16 (D1)	**17 (D2)**	**18 (D3)**	**19 (D4)**	**20 (D5)**
Purchases (Alternatives)	**16 (A1)**	32	32	32	32	32
	17 (A2)	29	34	34	34	34
	18 (A3)	26	31	36	36	36
	19 (A4)	23	28	33	38	38
	20 (A5)	20	25	30	35	40

Opportunity Loss (Regret) Matrix:

Purchase quantity \ Demand	16	17	18	19	20
16	0	2	4	6	8
17	3	0	2	4	6
18	6	3	0	2	4
19	9	6	3	0	2
20	12	9	6	3	0

Why are there two different types of opportunity loss?

If we purchase less than demand:

Opportunity loss = (5−3) × (shortage) = ₹ 2 × shortage

because we lose the ₹ 2 profit that could have been earned on each additional copy.

If we purchase more than demand:

Opportunity loss = ₹ 3 × (unsold copies)

because each unsold magazine costs ₹3 and has no salvage value.

2. Calculate EOL

The formula is EOL = ∑ P (D) × OL where D represents demand.

For purchasing 16 copies

EOL (16) = 0.10 (0) + 0.15 (2) + 0.20 (4) + 0.25 (6) + 0.30 (8) = 0 + 0.30 + 0.80 + 1.50 + 2.40 = ₹ 5.00

For purchasing 17 copies

EOL (17) = 0.10 (3) + 0.15 (0) + 0.20 (2) + 0.25 (4) + 0.30 (6) = 0.30 + 0 + 0.40 + 1.00 + 1.80 = ₹ 3.50

For purchasing 18 copies

EOL (18) = 0.10 (6) + 0.15 (3) + 0.20 (0) + 0.25 (2) + 0.30 (4) = 0.60 + 0.45 + 0 + 0.50 + 1.20 = **₹ 2.75**

For purchasing 19 copies

EOL (19) = 0.10 (9) + 0.15 (6) + 0.20 (3) + 0.25 (0) + 0.30 (2) = 0.90 + 0.90 + 0.60 + 0 + 0.60 = ₹ 3.00

For purchasing 20 copies

EOL (20) = 0.10 (12) + 0.15 (9) + 0.20 (6) + 0.25 (3) + 0.30 (0) = 1.20 + 1.35 + 1.20 + 0.75 + 0 = ₹ 4.50

3. EOL Summary

Purchase quantity	EOL (₹)
16	5.00
17	3.50
18	**2.75**
19	3.00
20	4.50

According to the **EOL criterion**, we select the purchase quantity having the **minimum expected opportunity loss**.

Min (EOL) = ₹ 2.75 which occurs when Purchase 18 copies

Final Answer: The newspaper boy should purchase 18 copies of the magazine.

The **minimum expected opportunity loss is ₹ 2.75 per day**.

Interpretation:

The decision is not based on the most probable demand alone. Although demand of **20 copies has the highest probability (0.30)**, purchasing 20 copies exposes the boy to greater losses when demand is lower. The EOL calculation balances the **loss from shortage** against the **loss from unsold magazines**, resulting in an order quantity of **18 copies**.

Problem 2: The sales of butter during 200 days have the following distribution:

Demand in kg	14	18	22	24	30	40
No. of days	10	20	20	80	40	30

The cost of butter is Rs. 80 per kg and it is sold at Rs. 120 per kg. Unsold stock is disposed of at the rate of Rs. 60 per kg the next day. Apply EMV criterion and EOL criterion to determine how much of butter is to be stocked at the beginning of each day.

Solution:

This is an **inventory decision problem under risk**. We can solve it by both **EMV (Expected Monetary Value)** and **EOL (Expected Opportunity Loss)** criteria.

Given Data: The sales/demand distribution over 200 days is:

Demand (kg)	**14**	**18**	**22**	**24**	**30**	**40**
Number of days	10	20	20	80	40	30
Probability	0.05	0.10	0.10	0.40	0.20	0.15

Check: 0.05 + 0.10 + 0.10 + 0.40 + 0.20 + 0.15 =1

Cost and selling information

- Purchase cost = ₹80/kg
- Selling price = ₹120/kg
- **Profit = Selling price - Purchase cost = ₹ 120 – ₹ 80 = ₹ 40 / kg**
- **Salvage value of unsold butter = ₹60 / kg**

Possible stock levels are: 14, 18, 22, 24, 30, 40 kg

Part A: EMV Criterion

Profit formula

If stock Q < D, everything is sold:

Profit = 120 Q – 80 Q = 40Q

If stock Q > D, then D kg is sold and Q−D kg are salvaged:

Profit = 120 D + 60 (Q−D) − 80Q

Simplifying, Profit = 60 D – 20 Q

1. Construct the Profit Table

Stock \ Demand	14	18	22	24	30	40
14 kg	**560**	560	560	560	560	560
18 kg	480	**720**	720	720	720	720
22 kg	400	640	**880**	880	880	880
24 kg	360	600	840	**960**	960	960
30 kg	240	480	720	840	**1200**	1200
40 kg	40	280	520	640	1000	**1600**

All entries are in **₹**.

Suppose the retailer stocks **30 kg**, but demand is only **24 kg**.

- 24 kg sold at ₹120 = **₹2,880**
- 6 kg remains and is salvaged at ₹60 = **₹360**
- Cost of 30 kg = **₹2,400**

Therefore, **Profit** = 2880 + 360 – 2400 **= ₹ 840**

2. Calculate EMV

EMV = ∑ P (D) × Profit

Stock 14 kg

EMV (14) = 560 (0.05 + 0.10 + 0.10 + 0.40 + 0.20 + 0.15) = ₹ 560

Stock 18 kg

EMV (18) = 480 (0.05) + 720 (0.10 + 0.10 + 0.40 + 0.20 + 0.15) = 24 + 684 = ₹ 708

Stock 22 kg

EMV (22) = 400 (0.05) + 640 (0.10) + 880 (0.10 + 0.40 + 0.20 + 0.15) = 20 + 64 + 748 = ₹ 832

Stock 24 kg

EMV (24) = 360 (0.05) + 600 (0.10) + 840 (0.10) + 960 (0.40 + 0.20 + 0.15) = 18 + 60 + 84 + 720 = ₹ 882

Stock 30 kg

EMV (30) = 240 (0.05) + 480 (0.10) + 720 (0.10) + 840 (0.40) + 1200 (0.20 + 0.15) = 12+48+72+336+420=**₹888**

Stock 40 kg

EMV (40) = 40 (0.05) + 280 (0.10) + 520 (0.10) + 640 (0.40) + 1000 (0.20) + 1600 (0.15) = 2 + 28 + 52 + 256 + 200 + 240 = ₹778

EMV Summary

The maximum EMV is: **₹ 888**

Therefore, according to the **EMV criterion**: Stock 30 kg of butter per day

Part B: EOL Criterion

For EOL, first identify the **maximum profit for each demand level**.

Demand	14 kg	18 kg	22 kg	24 kg	30 kg	40 kg
Maximum possible profit	₹ 560	₹ 720	₹ 880	₹ 960	₹ 1200	₹ 1600

The opportunity loss is: OL=Maximum profit for the demand−Actual profit

Opportunity Loss Table

Stock / Demand	**14**	**18**	**22**	**24**	**30**	**40**
14 kg	0	160	320	400	640	1040
18 kg	80	0	160	240	480	880
22 kg	160	80	0	80	320	720
24 kg	200	120	40	0	240	640
30 kg	320	240	160	120	0	400
40 kg	520	440	360	320	200	0

Calculate EOL: EOL = ∑P(D) × OL

For 14 kg

EOL (14) = 0 (0.05) + 160 (0.10) + 320 (0.10) + 400 (0.40) + 640 (0.20) + 1040 (0.15) = ₹ 492

For 18 kg

EOL (18) = 80 (0.05) + 0 (0.10) + 160 (0.10) + 240 (0.40) + 480 (0.20) + 880 (0.15) = ₹ 344

For 22 kg

EOL (22) = 160 (0.05) + 80 (0.10) + 0 (0.10) + 80 (0.40) + 320 (0.20) + 720 (0.15) = ₹ 220

For 24 kg

EOL (24) = 200 (0.05) + 120 (0.10) + 40 (0.10) + 0 (0.40) + 240 (0.20) + 640 (0.15) = ₹ 170

For 30 kg

EOL (30) = 320 (0.05) + 240 (0.10) + 160 (0.10) + 120 (0.40) + 0 (0.20) + 400 (0.15) = **₹ 164**

For 40 kg

EOL (40) = 520 (0.05) + 440 (0.10) + 360 (0.10) + 320 (0.40) + 200 (0.20) + 0 (0.15) = ₹ 274

EOL Summary

Stock (kg)	14 kg	18 kg	22 kg	24 kg	30 kg	40 kg
EOL (₹)	₹ 492	₹ 344	₹ 220	₹ 170	**₹ 164**	₹ 274

The minimum EOL is: ₹164 which occurs at **30 kg**.

Final Answer

Using EMV:

Optimal stock = 30 kg Maximum EMV = ₹ 888

Using EOL:

Optimal stock = 30 kg Minimum EOL = ₹ 164

Thus, **both EMV and EOL criteria give the same decision**:

Stock 30 kg of butter at the beginning of each day.

Useful relationship

This example also demonstrates the important relationship:

Minimum EOL = Expected Value with Perfect Information − Maximum EMV

Here, EVPI = 888 + 164 = ₹ 1052

So, the **Expected Value of Perfect Information (EVPI)** is ₹ 1,052 per day.

PART C: APPLICATIONS IN ENGINEERING AND COMPUTING

Applications in Engineering

Decision theory is widely used in engineering.

1. Machine selection

An engineer must select among different machines based on:

- Production demand
- Operating cost
- Maintenance cost
- Machine capacity

2. Plant capacity planning

A company must decide:

- Small plant
- Medium plant
- Large plant

based on future demand.

3. Reliability engineering

Engineers may decide whether to:

- Replace a component
- Repair a component
- Continue operating it

based on failure probabilities and costs.

4. Construction projects

A construction company may decide between different technologies depending on:

- Material prices
- Weather
- Demand
- Project duration

5. Energy systems

Power engineers can decide:

- How much generation capacity to install
- Which generation source to use
- How much reserve capacity to maintain

Applications in Computing and Computer Science

Decision theory has particularly important applications in computing.

1. Cloud computing

A company must decide whether to use:

- Small cloud instance
- Medium instance
- Large instance

depending on future workload.

States of nature:

- Low workload
- Medium workload
- High workload

Payoff:

- Profit
- Cost saving
- Response-time improvement

2. Server capacity planning

Suppose an online application expects uncertain traffic.

Possible states:

S1 = Low traffic

S2 = Medium traffic

S3 = High traffic

Actions:

A1 = Small server

A2 = Medium server

A3 = Large server

Decision theory helps determine the appropriate server configuration.

3. Machine Learning

Decision theory is closely related to machine learning.

A machine-learning system can estimate the probability of different outcomes and then select an action.

For example, a fraud-detection system may classify a transaction as:

- Genuine
- Suspicious

Actions:

- Approve
- Reject
- Send for manual review

The decision should consider the different costs of false positives and false negatives.

4. Cybersecurity

A security administrator may decide whether to:

- Ignore an alert
- Investigate
- Block the user/IP
- Isolate a system

States of nature may include:

- Genuine activity
- Cyberattack

The decision can be based on estimated probabilities and costs.

5. Software project selection

A software company may have several projects:

- AI project
- Mobile application
- Cloud application

Future market demand is uncertain.

Decision theory can help compare their possible returns.

6. Network management

Network administrators can decide whether to:

- Upgrade bandwidth
- Add servers
- Add redundancy
- Maintain the current infrastructure

based on expected traffic and failure conditions.

Practice problems

1. A producer of boats has estimated the following distribution of demand for a particular kind of boat.

No of boat demanded:	0	1	2	3	4	5	6
Probability	0.14	0.27	0.27	0.18	0.09	0.04	0.01

   Each boat cost him Rs. 7,000 and he sells them for Rs. 10,000 each. Any boat that are left unsold at the end of the season must be disposed off for Rs. 6000 each. How many boats should be in stock so as to maximize his expected profit?
2. An investor is given the following investment alternatives and percentage rates of return.

Strategies	**State of nature (Market conditions)**		
	Low	**Medium**	**High**
Regular shares	7 %	10 %	15 %
Risky shares	-10 %	12 %	25 %
Property	-12 %	18 %	30 %

   Over the past 300 days, 150 days have been medium market Conditions and 60 days have had high market increases. On the basis of these data, state the optimum investment strategy for the investment.
3. A company manufactures goods for a market in which the technology of the products is changing rapidly. The research and development department has produced a new product which to have potential for commercial exploitation. Further, Rs. 60000, is required or development testing.

   The company has 100 customers and each customer might purchase, at the most, one unit of the product. Market research suggests a selling price of Rs. 6,000 for each unit with total variable costs of manufacture and selling estimate are at Rs. 2000 for each unit.

   As a result of previous experiment of this type of market, it has been possible to derive a probability distribution relating to the proportions of customers who will buy the product as follows:

Proportions of customers:	0.04	0.08	0.12	0.16	0.20
Probability:	0.10	0.10	0.20	0.40	0.20

Determine the expected opportunity losses, given no information and to state whether, or not, the company should develop the product.

4. A milkman buys milk at Rs. 50 per litre and sells for Rs. 80 per litre. Unsold milk has to be thrown away. The daily demand in litres has the following probabilistic distribution:

Litres:	46	48	50	52	54	56	58	60	62	64
Probability:	0.01	0.03	0.06	0.1	0.2	0.25	0.15	0.1	0.05	0.05

   If each day's demand is independent of previous day. How many litres should be ordered every day?

5. A certain output is manufactured at Rs. 2 and sold at Rs. 4 per unit. The product is such that if it is produced but not sold during a week's time, it becomes worthless. The weekly sales record in the past are as follows:

Demand per week:	20	25	40	60
Number of weeks:	5	15	25	5

   Suggest the optimal act which should be taken by the manufacturer of the output.

6. A manufacturer manufactures a product, of which the principal ingredient is a chemical X. At the moment, the manufacturer spends Rs 1,000 per year on supply of X, but there is a possibility that the price may soon increase to four times its present figure because of a worldwide shortage of the chemical. There is another chemical Y, which the manufacturer could use in conjunction with a third chemical Z, in order to give the same effect as chemical X. Chemicals Y and Z would together cost the manufacturer Rs 3,000 per year, but their prices are unlikely to rise. What action should the manufacturer take? Apply the maximin and minimax criteria for decision-making and give two sets of solutions. If the coefficient of optimism is 0.4, then find the course of action that minimizes the cost.